Dogan, A. H.

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Education

Master of Science in Computer Engineering — 2022 – 10 Jan 2025
Middle East Technical University (METU) — *Ankara, Turkey*

Bachelor of Science in Computer Engineering — 2018 – 17 Jul 2022
Middle East Technical University (METU) — *Ankara, Turkey*

Experience

Research Intern — Dec 2024 – Mar 2025
Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich — *Zürich, Switzerland*

- Advanced research on continuous cybersickness detection in virtual reality environments using EEG-based multitaper spectrum estimation. Pioneered novel deep learning architectures achieving 30% performance improvement over baseline methods in real-time cybersickness prediction from brain signals, with potential applications in adaptive VR systems and user experience optimization.

Graduate Research Assistant — Feb 2022 – Present
Image Processing and Pattern Recognition Laboratory (ImageLab), METU — *Ankara, Turkey*

- Designed and implemented bucketed ranking loss with Sinkhorn-Knopp algorithm integration for DETR (Detection Transformer) architecture
- Achieved significant performance improvements in end-to-end object detection training through optimal transport-based loss formulations
- Published first-author paper at ECCV 2024, a premier computer vision conference (Acceptance Rate: ~27%)

Undergraduate Research Intern — Jun 2021 – Sep 2021
Cyber-Physical Systems Laboratory, University of California, Irvine — *Irvine, California, USA*

- Developed sophisticated multi-modal deep learning systems for cardiac and sleep abnormality detection from physiological signals. Designed novel architectures integrating time-series analysis with spectral feature extraction for beat-by-beat cardiac arrhythmia classification, achieving competitive performance in international challenges. Published research findings at Computing in Cardiology (CinC) 2021.

Publications

Dogan, A. H., Demirel, B. U., Holz, C. (2026). Frequency-Weighted Neural Kalman Filters. ***ICRA 2026***.

Demirel, B. U, **Dogan, A. H.**, Rossie, J., Möbus M., Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using eeg-based multitaper spectrum estimation. ***IEEE/TVCG 2025***.

Yavuz, F., Cam B.C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., Akbas, E. (2024). Bucketed Ranking Loss for Efficient Ranking-based Training of Object Detectors. ***ECCV 2024***.

Amin M. A., **Dogan, A. H.**, Kuru E. S., Sever Y., Angin P. (2024). Misuse Detection and Response for Orchestrated Microservices Based Software. ***AINA 2024***.

Demirel, B. U., **Dogan, A. H.**, Al-Faruque, M. A. (2021). Two Might Do: A Beat-by-Beat Classification of Cardiac Abnormalities using Deep Learning and Domain-Specific Features. ***CinC 2021***.

Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., Karagoz, P. (2021). Explainability in Irony Detection. ***DaWaK 2021***.

Sever Y., **Dogan, A. H.** (2023). A Kubernetes dataset for misuse detection. ***ITU Journal 2023***.

Sever, Y., Ekinci, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., Angin, P. (Sep 2022). An Empirical Analysis of IDS Approaches in Container Security. ***IEEE/SRMC 2022***.

Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, S. T., Tunel, B. M. (2024). Enhancing Underground Built Heritage Analysis with Text Mining: A Case Study on Cappadocia. ***Book Chapter 2024***.

Dogan, A. H., Dogan, A. (Jun 2021). An Assembled Deep Learning Approach for Flow Field Prediction. .

Honors & Awards

TUBITAK 2247 C STAR Scholarship — 2022-Present
The Scientific and Technological Research Institution of Turkey (TÜBİTAK)

Best Paper Award — September 2022
IEEE/SRMC'22

TUBITAK 2247 C STAR Scholarship — 2021-2022
The Scientific and Technological Research Institution of Turkey (TÜBİTAK)

TUBITAK 2247 C STAR Scholarship — 2020-2021
The Scientific and Technological Research Institution of Turkey (TÜBİTAK)

Technical Skills

Skills: Python/PyTorch, Variational and Generative Modeling, Computer Vision, Statistical Machine Learning, Combinatorial Optimization, Reinforcement Learning